Alberta's Two Maps: What the Audit Found

A plain-language summary of the 2025–26 Electoral Boundary Commission forensic audit. v0.21 — April 25, 2026 — Full monograph

An Airdrie resident pulls up the new electoral map on her phone. Her street goes to one riding. Two blocks over goes to another. Under the minority commission proposal, her city of 84,000 is cut into four pieces, each quarter attached to a different surrounding region. Nothing about Airdrie changed. Only the map did.

That map — and the second map produced by the same commission, and the legislative committee now drafting a third — is what this audit is about.

The short version

Alberta's Electoral Boundary Commission finished its work on March 23, 2026 and could not agree. Three commissioners produced one map; the other two produced a different one. Both are legal under the *Electoral Boundaries Commission Act*. This audit measured both using the same methods, applied identically. Three findings stand out.

1. **The two maps diverge systematically across six structural measures** — population spread, voter concentration, city fragmentation, municipal-boundary anchoring, map shape, and commission-chair-flagged anomalies. None of the six requires election results to calculate.
2. **The direction runs against the places where the governing party is weakest.** Communities most affected by the structural differences in the minority map — Calgary's northwest quadrant, the City of Airdrie, urban areas with established municipal boundaries — are the communities where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.
3. **The process now promoting the minority map departs from Canadian practice.** On April 16, the government set both commission maps aside and assigned redistricting to a five-MLA committee, three from the governing UCP. Most Canadian provinces either require the assembly to consider the commission report first or give commission maps automatic effect unless overridden. Alberta does neither.

The process is its own finding, separate from the maps.

The April 16 decision

The *Electoral Boundaries Commission Act* lets the government refer the commission's recommendations to a legislative committee. On April 16, the government did this. The new committee — chaired by Brandon Lunty, UCP MLA for Leduc-Beaumont — has five members. Three from the governing UCP, one from the NDP, one from the Alberta Party. It must report by November 2, 2026 with a 91-seat map.

The referral followed the letter of the Act. It also confirms Alberta's outlier status among Canadian provinces in two specific ways. Most provinces require the legislative assembly to consider a commission report before any referral can take effect; Alberta does not. Most provincial commissions are chaired by a sitting or retired judge whose tenure is protected; Alberta's chair is a public appointee who can be replaced mid-mandate by a simple cabinet decision.

Three Canadian comparator cases inform what "departing from norm" looks like in practice. In Quebec 1992 the National Assembly debated and amended a commission report rather than replacing the drafting process; the resulting map followed the commission's structure with targeted changes documented in committee transcripts. In Ontario 1996 a Conservative government collapsed 130 ridings to 103 by adopting the federal map, a structural rather than partisan change with bipartisan committee participation. In British Columbia 2008 a tied commission saw both reports tabled in the legislature with no government substitution; the assembly chose between them on a free vote. None of the three substituted a government-chaired drafting body for an independent commission's drafting process during the same redistribution cycle, which is what Alberta's April 16 motion did.

Three things the minority map does differently

It splits the City of Airdrie into four pieces. The law caps each electoral division at one-and-a-quarter times the provincial average, so Airdrie needs at least two divisions. The majority map gives it two. The minority map gives it four — North Airdrie attached to Olds–Three Hills–Didsbury, East Airdrie to Calgary-Falconridge–Conrich, West Airdrie to Cochrane-Springbank, and the urban core attached to Calgary-Foothills. An Airdrie resident with a question for her MLA has to know which quarter of the city she lives in before she can identify which legislative office to call. Her neighbours two blocks away will give her three different answers depending on which quarter they live in.

SIDEBAR — WHY AIRDRIE MATTERS

Population 84,000. Larger than Red Deer. One council, one tax bill, one school division.

Majority map: 2 divisions. Minority map: 4 — each quarter attached to a different surrounding region.

Both maps are legal. The four-way split is a choice.

It ignores municipal boundaries. When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends — the property-tax line, the school-division line, the local-election ward line, and the provincial-election line all coincide. The majority map traces municipal boundaries on 71 out of every 100 kilometres of its border. The minority map traces them on only 15 out of every 100. This number has nothing to do with which party benefits — it is measured from publicly available maps, not election results. Statistics Canada's Census Subdivision (CSD) layer is the survey-grade reference geography that any Canadian map-maker inherits for free. Drawing 85% of a province's electoral border off-reference is, in the Canadian commission literature, a deliberate methodological choice that requires explicit justification. The minority report offers none.

One area of Calgary is carved up to concentrate NDP voters into larger-than-average divisions. In Calgary's northwest quadrant, the minority map's divisions average 11.5% above the province-wide population — versus 2.8% on the majority. The same geographic zone, drawn by the same commission under the same constraints, produces districts a quarter larger on one map than on the other. Concentrating voters into oversized divisions dilutes their weight per ballot; redistricting analysts call it *packing* (see vocabulary sidebar).

Two further structural measures show the same pattern: the minority map's overall population spread is 48% wider than the majority's (median absolute deviation 4,707 vs 3,180), and the commission chair flagged three boundaries on the minority map as geographically anomalous (Rocky Mountain House–Banff Park's extension into uninhabited national park land; the Calgary-Nolan Hill–Cochrane lasso-shaped corridor; the Olds–Three Hills–Didsbury reach into north Airdrie). The majority received zero such flags from the same chair.

What the audit found across five measures

The table compares the two maps. None of the top six rows uses election results.

SIDEBAR — VOCABULARY

EFFICIENCY GAP (EG).

A wasted-vote count comparing the two parties. Above 7% in US case law has been argued (and rejected) as a gerrymander threshold. The Alberta-calibrated threshold is 4.37% (see methodology).

MEAN-MEDIAN.

The gap between the average and median district vote share. A persistent skew suggests one party's votes are systematically packed.

DECLINATION.

Treats the seats-votes relationship as an angle. Asymmetric angles flag districts

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
How evenly distributed the populations are	More even	48% more uneven
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Borders anchored to Statistics Canada census boundaries	80%	17%

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
Boundaries flagged by the commission chair	0	3
Efficiency gap (a partisan-fairness measure)	−1.3%	−2.7%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — *pre-registered pass*

The bottom two rows depend on election results and are less certain. Both efficiency-gap numbers favour the UCP relative to a perfectly neutral outcome — which is expected in Alberta, where concentrated urban NDP support produces a structural conservative tilt under any map. Both numbers fall well below any published gerrymander threshold, including the four Alberta-calibrated thresholds the audit derived (see methodology).

The last row is where the minority map looks *better*: zero packed-cracked adjacency pairs, against three on the majority and three on the current 2019 map — a pre-registered pass.

How the audit measures gerrymandering

A gerrymander is a deliberate boundary choice that advantages one political interest over others, beyond what the statutory constraints require. Two methodological problems make this hard to measure honestly.

The first is that the redistricting problem is mathematically *NP-hard*: there is no single correct Alberta map. Any map satisfying the *Electoral Boundaries Commission Act* — population deviation within ±25%, contiguity, compactness, community-of-interest preservation, Indigenous effective representation, and public-hearing input — is one of an astronomical family of legal maps. Fairness in this domain is not a discovery; it is a choice among many legitimate alternatives.

The second is that intent cannot be observed from a map. A commissioner could draw a partisan-skewed map either deliberately or by unlucky chance among the legal alternatives. The audit cannot read minds. What it can do is measure *effects* — how the map a commission produced compares to the family of maps a non-partisan procedure would have drawn.

The audit handles both problems with three layered tools.

A 100,000-map computer simulation. Using a redistricting algorithm called *ReCom* (Recombination Markov Chain Monte Carlo), the audit generates 100,000 different legal Alberta maps from the same statutory constraints the commission worked under. Each simulated map is scored on the four standard partisan-bias metrics. The real maps' scores are then placed into this distribution: a real map at the 5th or 95th percentile is a statistical outlier; a map between the 25th and 75th percentile sits in the typical range. The minority sits at the 95th-percentile boundary on mean-median; the majority sits at the same boundary. The 2019 enacted map sits at the same boundary. This tells the audit what is geography (all three baselines tilt the same way) versus what is drawing (the minority's structural irregularities, where the majority and 2019 do not show the same pattern).

Four Alberta-calibrated thresholds for the efficiency-gap test. The famous 7% threshold from US case law was calibrated to US Congressional delegation sizes and never adopted by any court. The audit derives four Alberta-specific alternatives: a first-principles assembly-size sensitivity threshold (~2.2%), an EBCA statutory-proportional threshold (5%), an Alberta historical-swing threshold (5–9% provisional), and the strongest — the MCMC ensemble's 95th-percentile EG of 4.37%, drawn from Alberta's actual 2019 substrate under EBCA constraints. Under partial-coverage geometry (67 of 89 EDs measurable), both 2026 maps fell below all four thresholds. Under the audit's now-current full-coverage geometry (89 of 89, via 2019-Tier-A inheritance fill for the 33 EDs whose 2026 boundaries could not be directly reconstructed), both maps exceed the 4.37% threshold (majority 6.4%, minority 9.2%) and both score in the top 0.1% of a 250,000-plan computer simulation. Multi-method disagreement on this magnitude is itself a finding — see Lane 1 below.

A pre-registered structural-irregularity scorecard. Five tests, committed to in writing before any results were read: anchoring departure from Canadian comparator norm, population MAD widening, geographic-zone gap, chair-flagged anomaly count, and rationale-failure count under symmetric audit. A map crossing four of five qualifies as a structural-irregularity outlier under the audit's own pre-registered rule. The minority crosses five of five. The majority crosses zero of five.

The combination of these three tools is what produces the two-lane verdict in the next section. No single tool is dispositive. Directional consistency across all three is the audit's inferential standard.

One way to read the scorecard

Rows of measurements are hard to weigh against each other. One defensible synthesis is a four-category rubric weighted by the principles redistricting law and democratic theory treat as primary.

SCORECARD — ONE WAY TO READ THE TWO MAPS

A proposed synthesis with explicit weights — not a finding. Other weightings give different grades.

CATEGORY	WEIGHT	MAJORITY	MINORITY
Voter equality	25%	B	**D**
Community recognition	35%	B	**F**
Map coherence	25%	A	**D**
Partisan fairness	15%	B	**B**
OVERALL	—	**B (87/100)**	**F (47/100)**

Why these weights. Community recognition carries the most (35%) because *Reference re Saskatchewan* — the Supreme Court decision defining what redistricting is for — holds that effective representation requires districts that communities can identify as their own. The Act's ±25% population band grounds voter equality at 25%. Map coherence (25%) reflects the commission's own standard: the chair flagged specific boundaries as anomalous, and those flags matter. Partisan fairness is weighted lowest (15%) because both maps fall below any published gerrymander threshold, and the direction of the partisan-fairness finding reverses depending on which measurement method you use.

What produces the minority's F. On community recognition (10/35 under these weights), 14.5% municipal anchoring means 85 out of every 100 kilometres of the minority map's border cuts through territory voters do not recognize from any other civic map — not their school division boundary, not their property-tax bill, not their local election ward. The majority's equivalent is 71%. The Airdrie four-way split compounds the effect — residents must navigate four legislative offices, each primarily identified with somewhere else. Three chair-flagged anomalies — a lasso-shaped corridor, a boundary extension through uninhabited national park land, and a district named for communities together smaller than the community it actually captures — pull the map-coherence score down further.

Where the maps score comparably. Under the lowest-weighted category — partisan fairness — the two maps are effectively tied, and neither shows a serious problem by any published standard.

Was either map a gerrymander?

The honest answer needs two lanes.

LANE 1 — THE PARTISAN-FAIRNESS NUMBERS

Where the two 2026 maps land against the Alberta-derived 4.37% threshold (the strongest of four Alberta-calibrated alternatives — see methodology), under the audit's full-coverage geometry (89 of 89 EDs measurable):

MAP	EFFICIENCY GAP	ABOVE ALBERTA 4.37%?	PERCENTILE VS 250K COMPUTER SIMULATION
Majority 2026	+6.4%	**Yes — 2.0 pp over**	**p99.9 (UCP-favoured tail)**
Minority 2026	+9.2%	**Yes — 4.8 pp over**	**p100 (UCP-favoured tail)**

On Lane 1, both maps now exceed the Alberta-derived gerrymander threshold. *This is a recent and consequential update. The earlier reading — that both maps were sub-threshold — used partial-coverage geometry (only 67 of 89 EDs) which under-counted votes in rural-UCP regions. The full-coverage 89-of-89 geometry the audit now publishes (built via 2019-Tier-A inheritance for the 21 majority + 12 minority EDs whose 2026 boundaries the commission's published maps did not allow direct reconstruction of) shifts the EG reading from sub-threshold to outlier on both maps.*

The honest framing: under partial-coverage attribution, neither map is a gerrymander on EG. Under full-coverage attribution, both are. The methods paper formalises this kind of cross-method disagreement; the audit reports both readings rather than picking one as authoritative. Until Elections Alberta releases the official 2026 shapefiles, the truth lies between the two readings.

Lane 2 — the structural pattern

Partisan-fairness numbers are not the only way to spot engineering. The audit applied nine further tests of *how the map was drawn* — all measurable from public maps, none dependent on election results.

TEST	MAJORITY MAP	MINORITY MAP
Border follows existing municipal lines (70–85% Canadian norm)	71% — within norm	**15% — 4.9× below norm**
Population spread (tighter is better)	3,180	**4,707 — 48% wider**
NW Calgary population concentration above average	0.4%	**12.2%**
Boundaries flagged by the commission's own chair	0	**3**
Airdrie split (constraint minimum: 2)	2 pieces	**4 pieces**
Minority's published reasons checked symmetrically	n/a	**6 of 7 fail**

TEST	MAJORITY MAP	MINORITY MAP
Contested choices with documented public-submission support	n/a	**2 have none**
Pre-registered structural-irregularity count (4 of 5 = outlier)	0 of 5	**5 of 5**
Mean-median percentile vs computer simulation	At the 95th-percentile boundary	At the 95th-percentile boundary

On Lane 2, the majority crosses zero structural thresholds. The minority crosses every one of them by a wide margin.

The six-of-seven rationale failures

The minority report includes published rationales for its seven most contested redraws. Symmetric audit checks each rationale against the data the rationale itself invokes. Six of seven fail.

- **Airdrie 4-way split** is justified by "rural and urban character differs across Airdrie's quadrants." Statistics Canada population profiles show Airdrie's four quadrants are within 8% of each other on every demographic measure the commission considers. Rationale fails.
- **Cochrane attached to Calgary-Foothills via the Nolan Hill lasso** is justified by "commuter community of interest." Journey-to-work data shows Cochrane's modal commuter destination is Cochrane itself; among out-of-town commuters, Calgary-Centre is named more often than Foothills. Rationale fails on the commute claim and on the geography claim.
- **Rocky Mountain House extended into Banff National Park** is justified by "ranching community of interest." The extended polygon contains zero year-round residents and zero working ranches. Rationale fails as stated.
- **Olds-Three Hills-Didsbury reaching into North Airdrie** is justified by "linked agricultural service centres." North Airdrie is suburban; Olds and Didsbury are rural service centres. The connection is geographic adjacency, not COI. Rationale fails.
- **Lethbridge split between Lethbridge-East and Taber-Warner** is justified by "matches federal boundary." The federal boundary it claims to match was retired in 2013. Rationale fails as stated.
- **Red Deer attached to Sylvan Lake** is justified by "shared school division." The shared school division contains 8% of Red Deer's school-aged population. Rationale fails on magnitude.
- **St. Albert-Sturgeon** is justified by "population balancing on Edmonton-corridor constraints." This rationale survives audit — the population deviation constraint, plus the Edmonton-corridor commute geography, leaves no other configuration that satisfies both the COI and the ±25% rule simultaneously. **Rationale stands.**

The six-failure / one-stand pattern matters. A symmetric audit that found 0/7 or 7/7 failures would suggest either no rationale problem or a methodology bias. A 6/7 pattern with the seventh being a constraint-forced configuration is the engineered/forced ratio that distinguishes deliberate boundary choice from constraint trade-off.

Verdict

	LANE 1: NUMBERS (FULL COVERAGE)	LANE 2: STRUCTURE
Majority 2026	EG +6.4% (over threshold)	not a gerrymander
Minority 2026	EG +9.2% (over threshold)	exhibits a gerrymander signature

Lane 1 verdict shifted with the move from partial to full coverage. Under partial coverage, neither map was over threshold; under full coverage, both are. The ENGINEERING DIFFERENCE that distinguishes the two maps lives in Lane 2, not Lane 1 — both maps inherit Alberta's structural rural-UCP advantage on EG; only the minority crosses every structural threshold by wide margins.

OPINION — THE EFFECTS, REPORTED PLAINLY.

The measurement work returns a coverage-sensitive result on Lane 1 (under partial-coverage attribution both maps were sub-threshold; under full-coverage attribution both exceed the Alberta-derived 4.37% EG threshold) and an unambiguous result on Lane 2 (the minority crosses every structural threshold by wide margins; the majority crosses none). Each individual structural signal has an innocent explanation available; the cumulative pattern is what the academic redistricting literature treats as a gerrymander signature. Whether the pattern's most parsimonious explanation is engineering, unlucky drafting, or some combination is the inference the reader is invited to weigh. The audit measures the effects; it does not claim to read the drafters' minds.

BOTTOM LINE:

Lane 1 finds both maps now above the Alberta-derived gerrymander threshold under full coverage (a recent and consequential update from earlier partial-coverage readings); Lane 2 finds only the minority crosses every structural threshold. The literature weights the cumulative structural pattern more heavily than any single magnitude metric; the monograph (§6.2) sets out the cross-method disagreement explicitly without picking one reading as authoritative.

How close did the minority map come?

- **On the partisan-fairness numbers (full coverage):** about 4.8 percentage points OVER the Alberta-calibrated 4.37% threshold (minority EG +9.2%). The majority is also over by 2.0 pp (+6.4%). Both maps land in the top 0.1% of a 250,000-plan computer simulation. Note that the partial-coverage reading put both maps under threshold; the full-coverage shift is itself a finding.
- **On the structural pattern:** already over every threshold the audit applies, by wide margins. The 4.9× anchoring departure from Canadian norm is the single largest threshold cross the audit measured.

The minority map maxes out the structural-irregularity scoring while staying inside the partisan-fairness numbers band.

SCORECARD — RETRACTION CONDITIONS

If any of these turn out to be true, the verdict above is wrong and gets retracted publicly within 30 days.

1. **A counter-map exists.** *Someone produces a legal Alberta map satisfying the minority's own community-of-interest reasons (Airdrie, Cochrane, Nolan Hill, Rocky Mountain House–Banff Park) and anchoring on municipal boundaries at majority-comparable rates. Open challenge — Issue #14 on the audit's GitHub repository.*
2. **The Lunty committee map looks like the minority.** *The November 91-seat map produces structural-irregularity counts in the same range — suggesting Alberta-specific drawing difficulty rather than minority-specific engineering.*
3. **A pre-2026 internal commission document surfaces.** *Showing the minority's choices were a deliberate response to documented community submissions rather than drafting choices.*
4. **The cross-election reversal extends to recent polling.** *The structural pattern would become harder to read as engineering if the partisan direction is genuinely vote-distribution-dependent.*

What this audit is now for

In policy terms, the verdicts above are moot. On April 16 the government set both commission maps aside; Alberta's 2027 map will come from the Lunty committee. Neither commission map is a candidate to become law.

The work itself is not moot. The commission audit calibrated Alberta-specific thresholds — structural-irregularity tests, anchoring comparators, a 100,000-plan computer ensemble — against three real Alberta maps (the 2019 enacted boundaries plus the two 2026 commission proposals, applied symmetrically). That framework will be re-pointed at the Lunty committee's map the day it is published, reported by the same standard you have just read.

The standard for November is the standard applied here: *skew toward neutrality wherever the constraints permit, document the departures, and defend them with evidence stronger than aesthetic preference.* The same scripts will answer, publicly, against the same thresholds — because we'll see if they don't.

SIDEBAR — TWO REMINDERS FOR THE NOVEMBER AUDIT

Alberta's geography will always favour conservatives. NDP voters concentrate in city cores; UCP voters spread across efficient-margin rural ridings. A neutrally-drawn Alberta map produces a UCP-favourable efficiency gap as a starting point — that's geography, not gerrymandering.

Many things count as gerrymandering, and not all show up in the partisan-fairness numbers. The minority map sat sub-threshold on every Alberta-calibrated EG line while crossing every structural threshold by a

What the audit does not claim

The audit does not claim the minority map was drawn to help the UCP win seats — intent cannot be proven from maps alone. It claims only that the differences are measurable, consistent, and not explained by Alberta's geography.

Both maps satisfy the *Electoral Boundaries Commission Act*.

The partisan-fairness numbers are sensitive to which election's results you use. Two different measurement methods give answers that point in opposite directions. The audit reports both, explains why they disagree, and does not pick one as correct. This is not evasion — it is an honest statement of the limits of what public data can tell you without the official map files, which Elections Alberta has not released.

The audit does not reach a constitutional conclusion. The *Reference re Saskatchewan* [1991] effective-representation framework is the standard a court would apply; whether the evidence here supports a challenge under that standard is for counsel and the courts to assess.

Behind the audit

This audit was produced by Will Conner, a fourth-year computer information systems student at Mount Royal University in Calgary. It is an independent research project with no funding from any political party or advocacy organization. The author is a UCP-disinclined Alberta voter; that prior is disclosed in the monograph's Author Disclosure block, along with the three findings retained in the paper that ran *against* the prior.

The audit used publicly available data: Elections Alberta's 2023 Statement of Vote (vote totals by voting area), Statistics Canada's 2021 Census (population, dissemination areas, census subdivisions), and the commission's 2026 final report (per-ED population tables, published rationales, chair-flag annotations).

All code is published in the GitHub repository and can be run independently by any reader. The pre-registered checklist that drove the audit was published before any 2026 results were measured; it is timestamped via Open Science Framework with embargoed release scheduled for 2026-11-02 (the day the Lunty committee must report).

The Elections Alberta shapefiles for the 2026 proposed boundaries have not been released. The audit reconstructs them from the commission's 600-DPI map images using a derived-provisional-geometry pipeline (eight stages, documented in monograph §4.1.5–§4.1.6). The reconstruction has documented ±500m perimeter precision and Tier-dependent area precision; every metric in this article that depends on geometry carries that caveat in the full monograph. A sunset clause binds the audit to recompute all geometry-dependent findings within 48 hours of any official shapefile release.

An AI assistant (Claude, made by Anthropic) helped draft and check the text, including this sentence. All factual claims were verified by the author against primary sources. The audit declares this openly because the same standard it applies to the commission — transparency about process — applies to the audit itself.

What you can do

1. **Read the full findings.** The monograph (report_academic.md) contains every number, every source, and every caveat. The retraction conditions tell you exactly what evidence would change each conclusion.
2. **Ask your MLA what the committee's process will be.** - Will the committee publish the evaluation criteria it uses *before* drawing a single line? - Will it release the range of maps it considered, not just the one it chose? - Will it publish any prompts or inputs given to AI tools?
3. **Request the official shapefiles.** Elections Alberta has not published the 2026 electoral division map files. Without these, some of this audit's calculations cannot be fully verified or corrected. A FOIP request would require their release.
4. **Share this report.** The full monograph is written for researchers and journalists. This report is written for everyone else. If you think the commission's process matters, share what you found here.

The full technical monograph, with all methods, caveats, and source citations, is at report_academic.md.